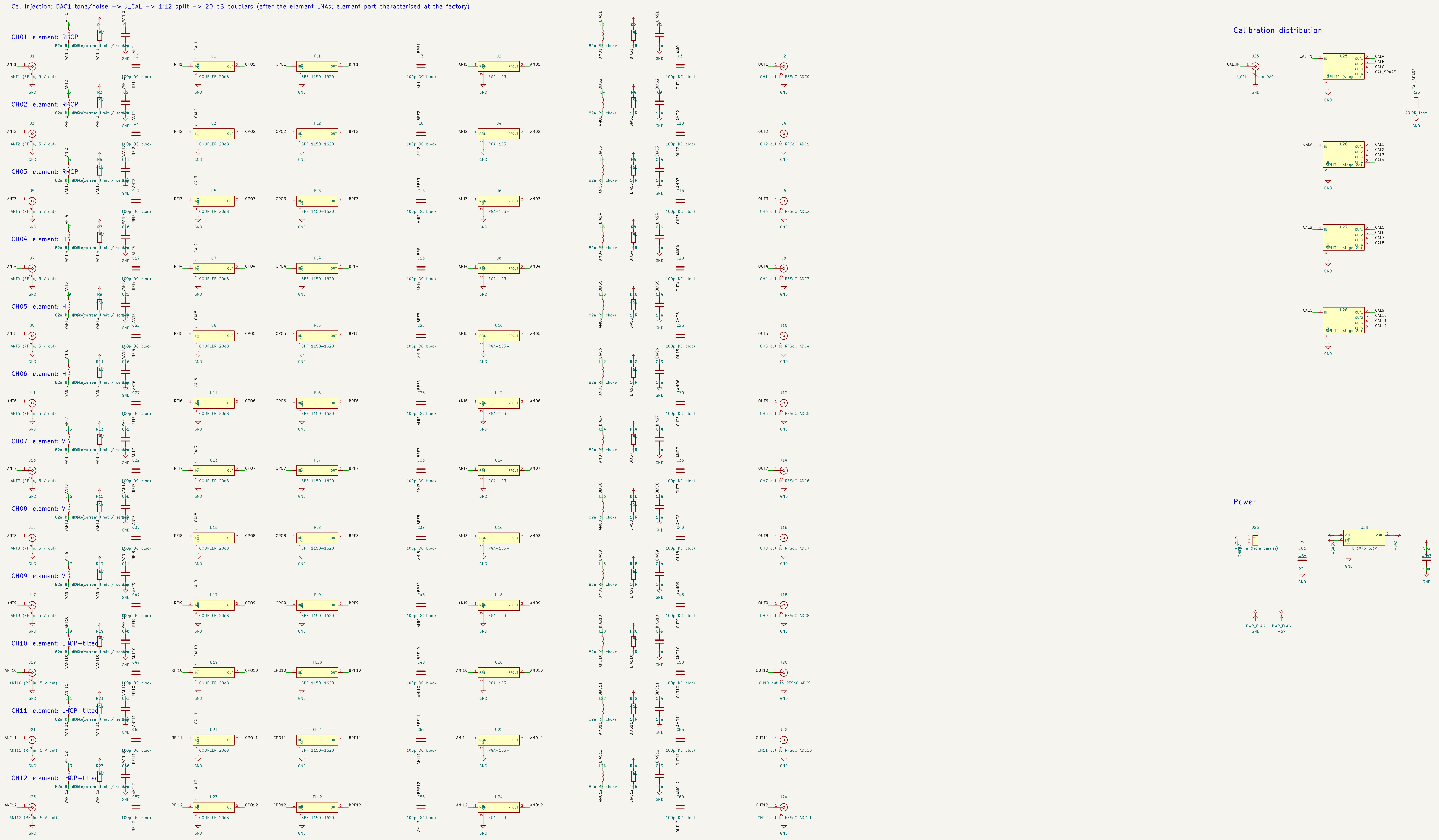
CRPA 12-channel RF front-end — one row per antenna element. Rev C: active elements (LNA at the antenna); this board: bias-tee, cal coupler, BPF, gain block.

Cal injection: DAC1 tone/noise -> J_CAL -> 1:12 split -> 20 dB couplers (after the element LNAs; element part characterised at the factory).


CH01 element: RHCP
CH02 element: RHCP
CH03 element: RHCP
CH04 element: H
CH05 element: H
CH06 element: H
CH07 element: V
CH08 element: V
CH09 element: V
CH10 element: LHCP-tilted
CH11 element: LHCP-tilted
CH12 element: LHCP-tilted
J1
ANT1 (RF in, 5 V out)
L1
82n RF choke
R1
10R (current limit / sense)
C1
10n
C2
100p DC block
U1
COUPLER 20dB
CAL1
FL1
BPF 1150–1620
C3
100p DC block
U2
PGA-103+
L2
82n RF choke
R2
10R
C4
10n
C5
100p DC block
J2
CH1 out to RFSoC ADC0
GND
+5V
Calibration distribution
J25
CAL_IN
J_CAL in from DAC1
U25
SPLIT4 (stage 1)
CALA
CALB
CALC
CAL_SPARE
R25
49.9R term
U26
SPLIT4 (stage 2a)
CAL1
CAL2
CAL3
CAL4
U27
SPLIT4 (stage 2b)
CAL5
CAL6
CAL7
CAL8
U28
SPLIT4 (stage 2c)
CAL9
CAL10
CAL11
CAL12
Power
J26
+5V in (from carrier)
C61
22u
U29
VIN
EN
VOUT
LT3045 3.3V
+3V3
C62
10u
PWR_FLAG
GND
PWR_FLAG
+5V

Part numbers marked VERIFY are candidates: check datasheet, pinout, availability
Generated by hardware/*/gen_*.py — edit the generator, not this file
GNSS anti-jam CRPA
Sheet: /
File: rf_frontend.kicad_sch
Title: CRPA 12-channel RF front-end
Size: A0
Date:
Rev: A
KiCad E.D.A. 10.0.3
Id: 1/1